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KidneyTox_v1.0 可实现小分子肾毒性可解释的人工智能预测

Sk Abdul Amin[1]✉、Supratik Kar[2]✉ 和Stefano Piotto[1]

药物引起的肾毒性仍然是肾功能障碍的主要原因，常常伴有严重甚至致命的后果。计算方法，尤其是人工智能（AI），通过提供可靠、经济且符合伦理的工具来评估药物引起的肾毒性，提供了一种有前景的替代方案，从而可能减少对动物实验的依赖。本研究的驱动因素有三个核心目标：（i）分析与药物引起肾毒性相关的化合物的化学空间；（ii）构建一个稳健的监督式机器学习（ML）分类模型，随后进行定量类比结构-活性关系（qRASAR）研究；以及（iii）开发一个名为“KidneyTox_v1.0”（https://kidneytoxv1.streamlit.app/）的开放访问、可解释人工智能（XAI）平台，用于肾毒性预测。除了提供预测之外，“KidneyTox_v1.0”还通过基于SHAP的互动瀑布图提供可解释性。

关键词 肾毒性，化学空间，指纹，机器学习，qRASAR

肾脏毒性或肾毒性在早期药物发现和开发中是一个重要的瓶颈，对临床前和临床失败以及上市后撤药有实质性贡献[1,2]。肾脏在过滤废物、重吸收必需物质和维持体内稳态方面的复杂生理功能，使其特别容易受到各种化学物质及其代谢产物的损伤[3]。肾毒性尤其难以通过计算方法预测，因为肾脏在过滤过程中会接触高浓度的药物和代谢产物，且毒性由多种机制驱动，如肾小管损伤、肾小球损伤、氧化应激和线粒体功能障碍，这与通常涉及更明确途径的肝毒性或心脏毒性不同。肾组织损伤可能导致急性肾损伤（AKI）或进展为慢性肾病（CKD），这些状况与患者的显著发病率和死亡率相关，并且

如今，包括机器学习（ML）、人工智能（AI）、定量构效关系（QSAR）、类比推断（RA）以及定量类比构效关系（qRASAR）在内的计算方法整合，已成为应对CKD挑战的强大范式[6,7]。这些方法利用日益增多的结构和毒性数据来建立预测模型，从而阐明化学物质（药物）与肾毒性之间的复杂相互作用。通过分析结构骨架、分子指纹及其他理化特征，计算工具可以帮助识别与肾毒性相关的结构警示和关键特征，从而指导开发具有更低流失率的更安全药物[8,9]。

不同研究团队的众多研究已成功采用人工智能/机器学习（AI/ML）和定量构效关系（QSAR）技术来预测各种毒理学终点，包括肾脏毒性[10–12]。早期的定量方法显示

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KidneyTox_v1.0 enables explainable artificial intelligence prediction of nephrotoxicity in small molecules

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Drug-induced nephrotoxicity remains a leading cause of kidney dysfunction, often with severe or even fatal outcomes. Computational approaches, in particular artificial intelligence (AI), offer a promising alternative by providing reliable, cost-effective, and ethically sound tools for assessing drug-induced nephrotoxicity. Thereby, potentially reducing reliance on animal testing. This study was driven by three core objectives: (i) to analyze the chemical space of compounds associated with drug-induced nephrotoxicity, (ii) to construct a robust supervised machine learning (ML) model for classification, followed by a quantitative Read-Across Structure-Activity Relationship (qRASAR) study, and (iii) to develop an open-access, eXplainable AI (XAI) platform named "KidneyTox_v1.0" (https://kidneytoxv1.streamlit.app/) for nephrotoxicity prediction. Beyond providing predictions, "KidneyTox_v1.0" offers interpretability through interactive SHAP-based waterfall plots, enabling both domain experts and non-experts to understand the contribution of molecular descriptors to toxicity outcomes. These modelling analyses will assist chemists in designing less nephrotoxic molecules in the future.

Keywords Nephrotoxicity, Chemical space, Fingerprint, Machine learning, qRASAR

Kidney toxicity or nephrotoxicity represents a significant bottleneck in the early-stage drug discovery and development, contributing substantively to preclinical and clinical failures and post-market withdrawals[1,2]. The intricate physiological role of kidney in filtering waste products, reabsorbing essential substances, and maintaining homeostasis makes it specifically susceptible to damage by various chemical entities and their metabolites[3]. Nephrotoxicity is especially difficult to predict computationally because kidneys face high drug and metabolite exposure during filtration, with toxicity driven by diverse mechanisms like tubular injury, glomerular damage, oxidative stress, and mitochondrial dysfunction, unlike hepatotoxicity or cardiotoxicity, which often involve more defined pathways. Damage to renal tissue can lead to acute kidney injury (AKI) or progress to chronic kidney disease (CKD), conditions that are associated with significant patient morbidity and mortality and that result in high attrition rates during drug development. Detecting and mitigating probable nephrotoxicity early in the drug discovery pipeline is paramount to minimizing attrition rates, reducing development costs, and ensuring patient safety[4]. Traditional methods for evaluating kidney toxicity, including in vitro cell assays and in vivo animal models, are often time-consuming, resource-intensive, and may not always accurately predict human response[5]. Consequently, there is a critical need for more efficient and predictive approaches to assess the nephrotoxic potential of novel drug candidates.

Now-a-days, the integration of computational approaches, including Machine Learning (ML), Artificial Intelligence (AI), Quantitative Structure-Activity Relationships (QSAR), Read-Across (RA), and quantitative Read-Across Structure-Activity Relationships (qRASAR), have emerged as a powerful paradigm to address the CKD challenges[6,7]. These approaches support the ever-increasing availability of structural and toxicity data to build predictive models that can elucidate the complex interactions between chemicals (drugs) and nephrotoxicity. By analyzing structural scaffolds, molecular fingerprints, along with other physicochemical features, computational tools can assist in identifying structural alerts and key features associated with kidney toxicity, guiding the design of safer drugs with a lower attrition rate[8,9].

Numerous studies by different research groups have successfully employed AI/ML, and QSAR techniques for predicting various toxicological endpoints, including kidney toxicity[10–12]. Early quantitative approaches showed

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当机器学习模型在策划的化学数据集上进行训练时，它们能够捕捉分子指纹与毒性终点之间的复杂线性和非线性关系。例如，研究人员已经采用从支持向量机（SVM）到随机森林（RF）以及梯度提升（GB）方法的多种技术，在预测性能上持续超过传统经验模型[13]。此外，通过整合基于片段的分析以识别毒性基团，即与肾毒性密切相关的特定分子亚结构，这些研究推动了我们对肾毒性分子框架的理解[14]。

风险评估方法，通常与定量构效关系（QSAR）结合使用，已被用于基于结构相似化合物已知的毒性预测目标化学品的毒性[15]。这些努力展示了计算方法在提供潜在肾毒性早期见解方面的潜力，可补充传统的实验方法。然而，在开发在不同化学空间和不同肾毒性机制中广泛适用的高精度、稳健的预测模型方面仍然存在挑战。高质量、经过整理的数据集的可用性也是一个挑战，这些数据集应同时包含“有毒”和“无毒”的反应。此外，使这些预测工具易于药物化学家和毒理学家访问和解读仍然是一个关键目标[16]。多年来，这类数据集的异质性和有限规模限制了预测模型的发展。

为应对这些挑战，我们开发了一种全面的计算方法和一个用户友好的基于网络的工具“KidneyTox_v1.0”，用于预测肾毒性。我们的研究利用了一个经过整理的数据集，其中包含565种小分子及其与肾毒性相关的实验确定的“有毒”和“无毒”反应。我们采用了一种多方面策略，包括：(i) 化学空间分析以了解数据集的分布和多样性；(ii) 基于人工智能/机器学习（AI/ML）的QSAR建模，随后使用qRASAR构建一个稳健的预测和解释模型；(iii) 开发一个开放访问的、可解释的人工智能（XAI）平台，用于肾毒性预测。

虽然所使用的机器学习模型的具体细节将在方法部分讨论，但该模型是基于其性能和可解释性而选择的。开发的预测模型已集成到“KidneyTox_v1.0”（https://github.com/Amincheminform/KidneyTox_v1.0）中，为用户提供了一个开源平台，以便快速获取预测结果。用户可以通过提供化学结构的SMILES字符串，或在直观的基于Streamlit的网页界面中直接绘制结构，轻松输入化学结构。除了简单的二元预测外，该平台还生成有用的可视化图表，包括适用性域（AD）分析、水fall图，这些图表说明主要描述符/特征对预测肾毒性的正负贡献。此外，用户还可以获得与其查询化学物质在建模化学空间中结构相似的化学物质的对比信息，以及它们各自的预测结果。

结果

一个药物诱导的肾毒性数据集（N_{Total} = 565）来源于文献[17,18]。该数据集包含565种化学结构多样的小分子，包括287种已报道在人类中诱导肾毒性的药物（“有毒”）以及278种已知没有肾毒性作用的药物（“无毒”），如补充材料的表S1所示。

化学空间分析

化学空间在化学、生物学和毒理学研究中都起着关键作用，尤其是在药物化学中。图1展示了所研究分子的六种分子属性（辛醇-水分配系数（LogP）、分子量（MW）、氢键受体（HBA）、氢键供体（HBD）、环系统总数（nRings）、可旋转键数量（nRB））的频率分布。LogP值最高的分子具有很强的亲脂性，值为9.16（化合物368）。相反，化合物479是最亲水的，LogP值为-20.60。数据集中所有分子的平均LogP值为1.81，表明大多数分子具有中等亲脂性。没有芳香环的分子有142个，具有一个芳香环的分子有155个，具有两个芳香环的分子有166个，具有三个芳香环的分子有72个。因此，平均而言，分子拥有一至两个芳香环。有19个分子w

有31个分子没有可旋转键，38个分子有一个可旋转键。平均而言，分子大约有3个环。整个数据集分子的平均HBA和HBD值分别为6.10和2.89。因此，大多数分子大约有6个氢键受体和3个氢键供体。整个数据集分子的平均拓扑极性表面积（TPSA）和分子量（MW）分别为113.53和416.87。按分子量计算，化合物176（MW = 4491.95）是最大的分子，而化合物119（MW = 59.04）是最小的分子。分子的平均分子量（416.87）表明大多数分子为小型到中型分子。分子性质的广泛范围，特别是LogP、分子量和TPSA，表明了化学空间的多样性，包括小型有机化合物以及高度柔性的结构。

指纹辅助支架多样性分析

进行了指纹辅助的骨架多样性分析（整合化学信息学和无监督机器学习方法），以获得对骨架多样性的见解。在这里，我们使用开源工具“Fasda v1.0”（https://github.com/Amincheminfom/Fasda v1）来执行指纹辅助的骨架多样性分析

that ML models, when trained on curated chemical datasets, can capture complex linear as well as nonlinear relationships between molecular fingerprints and toxicity endpoints. For example, researchers have employed techniques ranging from support vector machines (SVM) to random forests (RF) and gradient boosting (GB) methods, consistently achieving improved predictive performance over traditional empirical models[13]. Moreover, by integrating fragment-based analyses to identify toxicophores, specific molecular substructures that are strongly associated with kidney toxicity, these studies have advanced our understanding of the molecular frameworks underlying nephrotoxicity[14].

RA approaches, often integrated with QSAR, have been employed to predict the toxicity of a target chemical based on the known toxicity of structurally similar compounds[15]. These efforts have demonstrated the potential of computational methods to provide valuable early insights into potential kidney toxicity, complementing traditional experimental methods. However, challenges remain in developing highly accurate and robust predictive models that are broadly applicable across diverse chemical spaces and different mechanisms of nephrotoxicity. Availability of high-quality, curated datasets that encompass both ‘Toxic’ and ‘Non-toxic’ responses is another challenge. Furthermore, making these predictive tools easily accessible and interpretable for medicinal chemists and toxicologists remains a crucial goal[16]. For many years, the heterogeneity and limited size of such datasets have constrained the development of predictive models.

Addressing these challenges, we have developed a comprehensive computational approach and a user-friendly web-based tool, “KidneyTox_v1.0”, for predicting kidney toxicity. Our study utilized a curated dataset of 565 small molecules with experimentally determined ‘Toxic’ and ‘Non-toxic’ responses related to kidney toxicity. We employed a multi-faceted strategy involving (i) chemical space analysis to understand the distribution and diversity of our dataset, (ii) AI/ML-based QSAR modeling followed by qRASAR to build a robust predictive and interpretative model, (iii) development of an open-access, eXplainable AI (XAI) platform for nephrotoxicity prediction.

While specific details of the ML model employed will be discussed in the methods section, it was selected based on its performance and interpretability. The developed predictive model is integrated into the “KidneyTox_v1.0” (https://github.com/Amincheminform/KidneyTox_v1.0), providing an open-source platform for users to obtain rapid predictions. Users can easily input a chemical structure either by providing its SMILES string or by directly drawing the structure within the intuitive Streamlit-based web interface. Beyond a simple binary prediction, the platform generates insightful visualizations, including applicability domain (AD) analysis, waterfall plots that illustrate the positive and negative contributions of major descriptors/features towards the predicted nephrotoxicity. Furthermore, users receive comparative information on chemicals within the modeled chemical space that are structurally similar to their query chemicals, along with their respective predicted outcomes and accompanying waterfall plots. This comparative analysis helps contextualize the predicted nephrotoxicity of the query molecule within the landscape of known compounds, enhancing the interpretability and functionality of the predictions.

Results

A dataset of drug-induced nephrotoxicity (N_{Total} = 565) was sourced from literatures[17,18]. This dataset comprises 565 chemically diverse small molecules, including 287 drugs reported to induce nephrotoxicity in humans (‘Toxic’) and 278 drugs with no known nephrotoxic effects (‘Non-toxic’) as given in Table S1 of the Supplementary material.

Analysis of the chemical space

Chemical space plays a crucial role in both chemical, biological and toxicological research, particularly in pharmaceutical chemistry. Figure 1 illustrates the frequency distribution of six molecular properties (octanol-water partition coefficient (*LogP*), molecular weight (*MW*), hydrogen bond acceptors (*HBA*), hydrogen bond donors (*HBD*), total number of ring systems (*nRings*), number of rotatable bonds (*nRB*)) of the molecules investigated. The molecule with the highest *LogP* value is highly lipophilic with a value of 9.16 (compound 368). Conversely, compound 479 is the most hydrophilic, with a *LogP* value of -20.60. The average *LogP* value across all molecules in the dataset is 1.81, indicating that most molecules are moderately lipophilic. There are 142 molecules without aromatic rings, 155 molecules with one aromatic ring, 166 molecules with two aromatic rings, and 72 molecules with three aromatic rings. Hence, on average, molecules have one or two aromatic rings. There are 19 molecules with four aromatic rings, 8 molecules with five aromatic rings, 2 molecules (example: compounds 396 and 483) with six aromatic rings, and 1 molecule (example: compound 176) with nine aromatic rings. Moreover, compound 176 also exhibits 151 rotatable bonds. Molecules across the datasets typically have around 7 rotatable bonds, indicating moderate flexibility.

There are 31 molecules without any rotatable bond and 38 molecules with one rotatable bond. On average, molecules have about 3 rings. The average *HBA* and *HBD* values across all dataset molecules are 6.10 and 2.89, respectively. Hence, most molecules have around 6 hydrogen bond acceptors and around 3 hydrogen bond donors. The average topological polar surface area (*TPSA*) and *MW* values across all dataset molecules are 113.53 and 416.87, respectively. Compound 176 (*MW* = 4491.95) is the largest molecule by weight, whereas compound 119 (*MW* = 59.04) is the smallest molecule by weight. The average *MW* (416.87) of molecules indicates that most are small to medium-sized. The broad ranges of molecular properties, particularly *LogP*, *MW*, and *TPSA*, suggest a diverse chemical space, including small organic compounds, and highly flexible structures.

Fingerprint-assisted scaffold diversity analysis

The fingerprint-assisted scaffold diversity analysis (integrating cheminformatics and unsupervised ML methods) was performed to gain insights into the scaffold diversity[19]. Here, we used the open-access tool “Fasda_v1.0” (https://github.com/Amincheminfom/Fasda_v1) to perform fingerprint-assisted scaffold diversity analysis

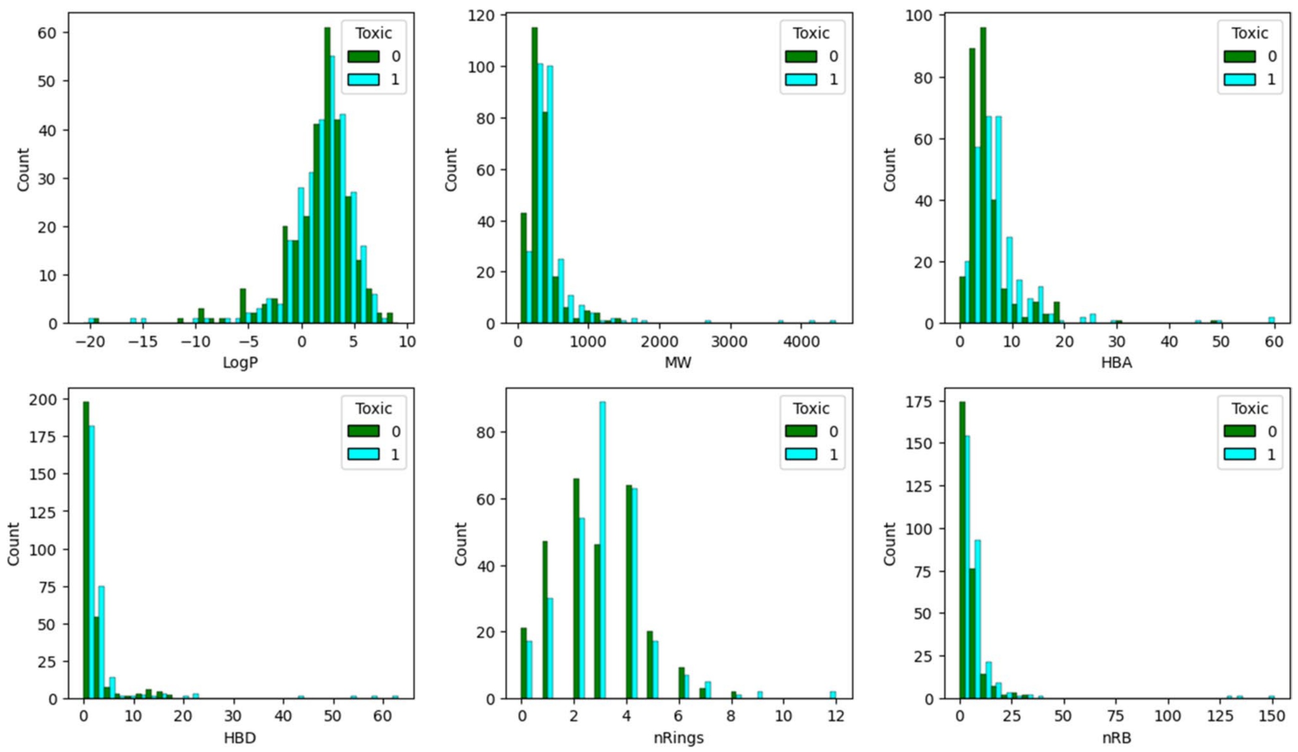

图1. 常见特征的直方图，例如 LogP、分子量 (MW)、氢键受体数 (HBA)、氢键供体数 (HBD)、环的数量 (nRings) 和可旋转键数 (nRB)。

与药物诱导肾毒性相关的化合物。首先，生成了ECFP_6 指纹，并使用Tanimoto系数计算两两相似性。随后应用主成分分析（PCA）进行二维可视化，然后进行K均值聚类（确定最佳聚类数的肘部图已在补充材料的图S1中提供），将分子划分为五个化学上相似的组。随后对独特骨架和单例骨架的分析突出了每个聚类内的整体结构多样性。从每个聚类中选取了两种代表性药物 [Perindoprilat（化合物38）、Rosuvastatin（化合物92）、Atazanavir（化合物156）、Acamprosate（化合物267）、Terlipressin（化合物333）、Eplerenone（化合物383）、Paroxetine（化合物414）、Telbivudine（化合物464）、Testosterone（化合物508）、Timolol（化合物558）]，如图2所示。

簇0以拥有大量化合物（M＝149）、Bemis-Murcko骨架（N＝126）和单一骨架（Ns＝108）而突出，表明其具有高度的结构多样性（表1）。高单一比率（0.857）也进一步表明该簇中的大多数骨架仅出现一次，反映了高度的结构多样性。簇1由52个不同的Bemis-Murcko骨架组成，单一比率为0.827。从表1可以看出，簇2有134个化合物，单一比率为0.876。此外，簇3（M＝50）和簇4（M＝139）分别显示出32和99个Bemis-Murcko骨架。簇3和簇4的单一比率分别为71.9%和89.9%。因此，所有簇的单一比率均高于70%，突显出整个数据集持续存在的高结构多样性。

机器学习研究

计算 o　f 描述　p扭矩器和数据 p再治疗

Mordred 描述符是使用内部“Fiore v1.0”平台计算的，特别是使用 Fiore FC（特征计算）工具（https://github.com/Amincheminfom/Fiore v1.0），在开发 AI/ML 模型之前进行计算。包含文本信息的列和常量列（在所有样本中值相同的列）被删除，以消除不必要的噪音并降低维度。值得注意的是，该数据集通过重复随机分层抽样被划分为 80% 的训练集和 20% 的测试集，并迭代了 50 种不同的随机划分。这 50 个独立的训练-测试划分被用于开发初步的随机森林分类器（RFC）模型，从而评估模型的稳健性，而不是依赖于单一划分。选择性能最好的划分，其结果显示在补充材料的图 S2 中。随后，使用所选训练集进行特征选择，然后使用优化后的 RFC 模型进行模型开发。

特征选择

特征选择是机器学习中一个关键的预处理步骤，它可以提升模型性能[22]。该步骤能够降低复杂性并保留核心信息，从而确保模型在训练过程中的稳定性。同时，计算了信息增益（或互信息）[23] 以识别与预测目标变量（毒性值）最相关的特征。最后，选择了重要性评分大于0.05的特征以提高准确性并

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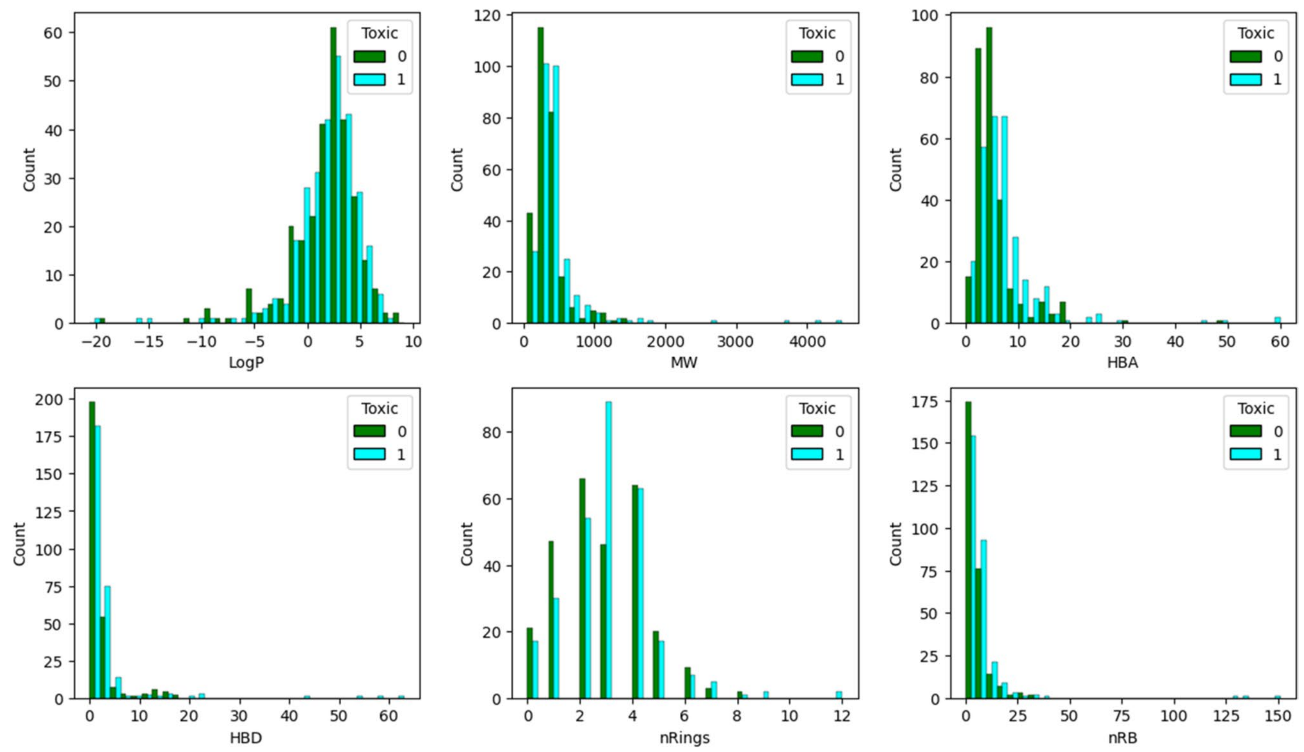

Fig. 1. Bin plots of common features such as *LogP*, *MW*, *HBA*, *HBD*, *nRings*, and *nRB*.

of compounds associated with drug-induced nephrotoxicity. First, *ECFP_6* fingerprints were generated, and pairwise similarities were computed using the Tanimoto coefficient. Principal Component Analysis (PCA) was applied for 2D visualization, followed by *K*-Means clustering (Elbow plot for determining the optimal number of clusters has been provided in Figure S1 of the Supplementary material) to partition the molecules into five chemically similar groups. Then the analysis of unique and singleton scaffolds highlighted overall structural diversity within each cluster. Two representative drugs [Perindoprilat (compound 38), Rosuvastatin (compound 92), Atazanavir (compound 156), Acamprosate (compound 267), Terlipressin (compound 333), Eplerenone (compound 383), Paroxetine (compound 414), Telbivudine (compound 464), Testosterone (compound 508), Timolol (compound 558)] from each cluster are provided in Fig. 2.

Cluster 0 stands out for having a high number of compounds (M = 149), Bemis-Murcko scaffolds (*N* = 126), and singleton scaffolds (Ns = 108), indicating a high degree of structural diversity (Table 1). The high singleton ratio (0.857) also reinforces that most scaffolds in this cluster occur only once, reflecting high structural diversity. Cluster 1 is expressed by 52 distinct Bemis-Murcko scaffolds with a singleton ratio of 0.827. It can be observed from Table 1 that cluster 2 has 134 compounds, with a singleton ratio of 0.876. Moreover, cluster 3 (M = 50) and cluster 4 (M = 139) exhibit Bemis-Murcko scaffolds of 32 and 99, respectively. The singleton ratio is 71.9% and 89.9% for cluster 3 and 4, respectively. Hence, all the clusters exhibit singleton ratios above 70%, highlighting consistently high structural diversity across the dataset.

Machine learning studies

Calculation of the descriptors and data pre-treatment

Mordred descriptors[20] were calculated by using *in house* “Fiore_v1.0” platform, in particular, *Fiore_FC* (Feature Calculation) tool (https://github.com/Amincheminfom/Fiore_v1.0) prior to developing AI/ML models. Columns containing text information, constant columns (those with identical values across all samples) were deleted to eliminate unnecessary noise and reduce dimensionality[21]. Notably, the dataset was divided into 80% training and 20% test sets using repeated random stratified sampling, iterating over 50 different random splits. These 50 independent train-test splits were used to develop preliminary Random Forest Classifier (RFC) models, allowing the assessment of model robustness rather than relying on a single partition. The best-performing split was selected, and the results are depicted in Figure S2 of the Supplementary material. Subsequently, feature selection was performed using the selected training set, followed by model development with RFC model optimized through Optuna-based hyperparameter tuning (https://optuna.org/).

Feature selection

Feature selection is a critical preprocessing step in ML that enhances model performance[22]. This step reduces complexity and preserves essential information, ensuring model stability during training. Meanwhile, information gain (or mutual information)[23] was calculated to identify features most relevant to predict the target variable (toxicity value). Finally, features with importance scores greater than 0.05 were selected to improve accuracy and

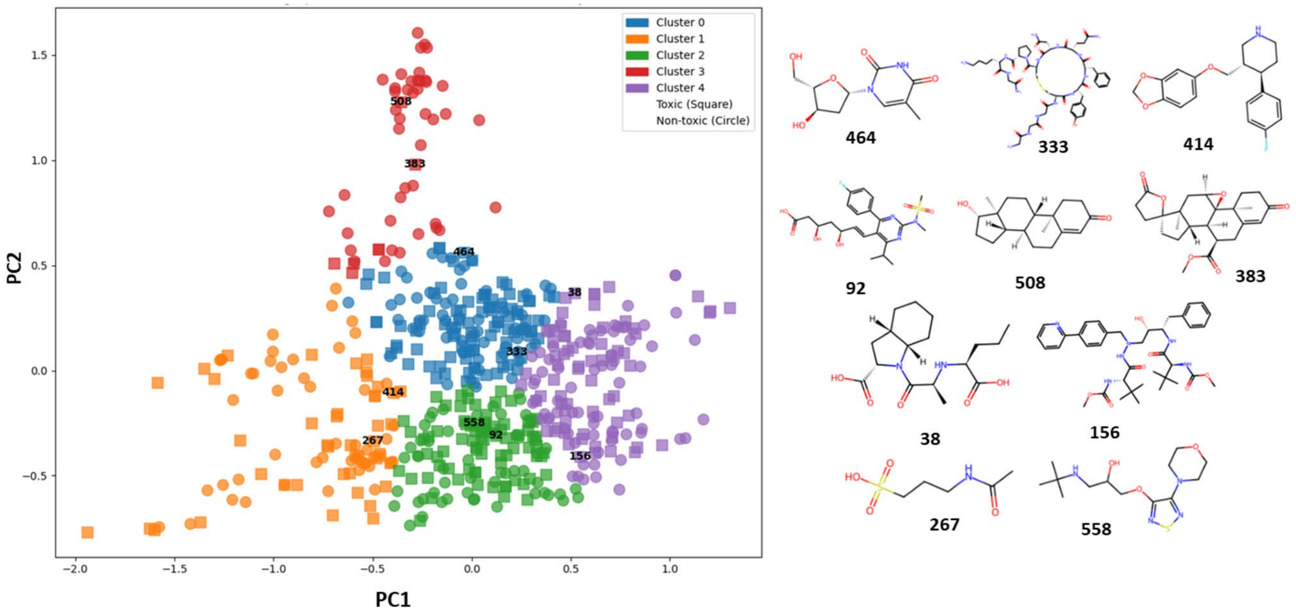

图2. PCA散点图突出了每个簇中的两个代表性药物 [培哚普利酯（化合物38）、瑞舒伐他汀（化合物92）、阿塔那韦（化合物156）、阿卡酯酸（化合物267）、特利普利辛（化合物333）、依普利酮（化合物383）、帕罗西汀（化合物414）、替比夫定（化合物464）、睾酮（化合物508）、丁螺莫尔（化合物558）]。

集群	M	N	Ns	单例比率
0	149	126	108	0.857
1	93	52	43	0.827
2	134	97	85	0.876
3	50	32	23	0.719
4	139	99	89	0.899

表 1. 药物诱导肾毒性数据集的骨架内容和多样性计算。化合物数量 (M)、独特骨架数量 (N)、单一骨架数量 (Ns)、单一骨架比率为单一骨架数量与骨架总数的比值。

降低计算成本。所选描述符共同提供了分子结构、性质和反应性的详细表征，使它们成为机器学习的强大工具。所选描述符的相关矩阵热图如图3A所示。

机器学习模型开发

在调优过程中确定的最优超参数如下：n_estimators = 60，max_depth = 24，min_samples_split（分裂一个节点所需的最小样本数）= 13，min_samples_leaf（叶节点必须具有的最小样本数）= 2。n_estimators 的值表明模型在使用 60 棵单独树时达到最佳性能。所选数值主要反映的是经验性能饱和，而非计算限制。所有这些超参数（n_estimators = 60，max_depth = 24，min_samples_split = 13，min_samples_leaf = 2）共同生成了一个判别能力强且复杂度合理的模型。训练集的性能指标包括准确率为 94.25%，精确率为 94.83%，召回率为 94.02%，F1 分数为 94.42%，凸显了 Optuna 优化的 RFC 模型的出色质量。此外，该模型每

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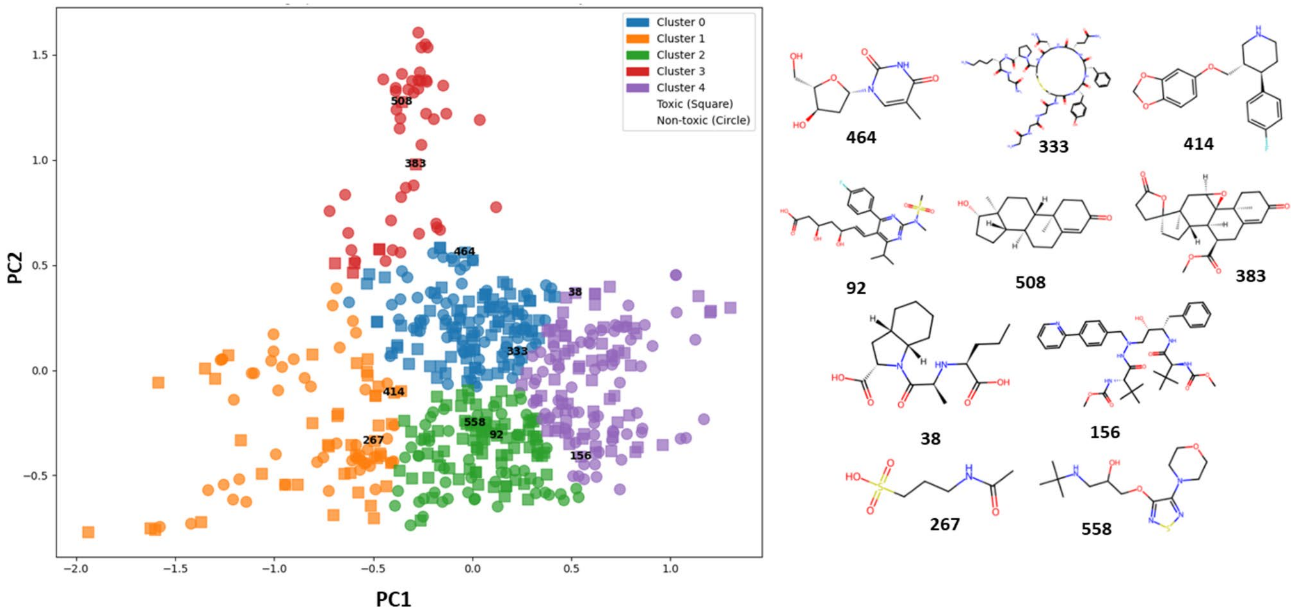

Fig. 2. The PCA scatter plot highlights two representative drugs [Perindoprilat (compound 38), Rosuvastatin (compound 92), Atazanavir (compound 156), Acamprosate (compound 267), Terlipressin (compound 333), Eplerenone (compound 383), Paroxetine (compound 414), Telbivudine (compound 464), Testosterone (compound 508), Timolol (compound 558)] from each cluster.

Cluster	M	N	Ns	Singleton Ratio
0	149	126	108	0.857
1	93	52	43	0.827
2	134	97	85	0.876
3	50	32	23	0.719
4	139	99	89	0.899

Table 1. Scaffold content and diversity calculation of the dataset of drug-induced nephrotoxicity. Number of compounds (M), Number of unique scaffolds (N), Number of singleton scaffolds (Ns), Singleton Ratio is the ratio of singleton scaffolds to total scaffolds.

reduce computational costs. The selected descriptors collectively provide a detailed representation of molecular structure, properties, and reactivity, making them powerful tools for ML. A heatmap of the correlation matix of the selected descriptors is depicted in Fig. 3A.

Machine learning model development

The optimal hyperparameters identified during the tuning process[24] are as follows: *n_estimators* = 60, *max_depth* = 24, *min_samples_split* (minimum number of samples required to split a node) = 13, *min_samples_leaf* (minimum number of samples that a leaf node must have) = 2. The value of *n_estimators* suggests that the model achieves optimal performance with 60 individual trees. The selected value primarily reflects empirical performance saturation rather than computational constraints. Together, all these hyperparameters (*n_estimators* = 60, *max_depth* = 24, *min_samples_split* = 13, *min_samples_leaf* = 2) produce a model with strong discriminative power while maintaining a reasonable level of complexity. The performance metrics of the training set include an Accuracy of 94.25%, Precision of 94.83%, Recall of 94.02%, and an F1 Score of 94.42%, highlighting the excellent quality of the Optuna-optimized RFC model. Additionally, this model performs well in the validation dataset, achieving a best accuracy score of 0.841, precision of 0.830, recall of 0.830 and F1 score of 0.830 on the test set, indicating its ability to capture generalizable patterns. Furthermore, ten alternative train-test splits were evaluated to develop additional RFC models. These splits yielded moderately lower performance (Table S2 of the Supplementary material) than the original RFC model, with accuracies ranging from 0.673 to 0.796 and corresponding variations in precision, recall, F1-scores, and optimized hyperparameters (e.g., Split 1: Accuracy = 0.673, F1 = 0.704; Split 6: Accuracy = 0.796, F1 = 0.803). These findings indicate that the original RFC model exhibits robust and the best predictive behavior across replicate splits. Finally, the plot of selected descriptors with their importance score as per the final RFC model is given in Figure S3A of the Supplementary material.

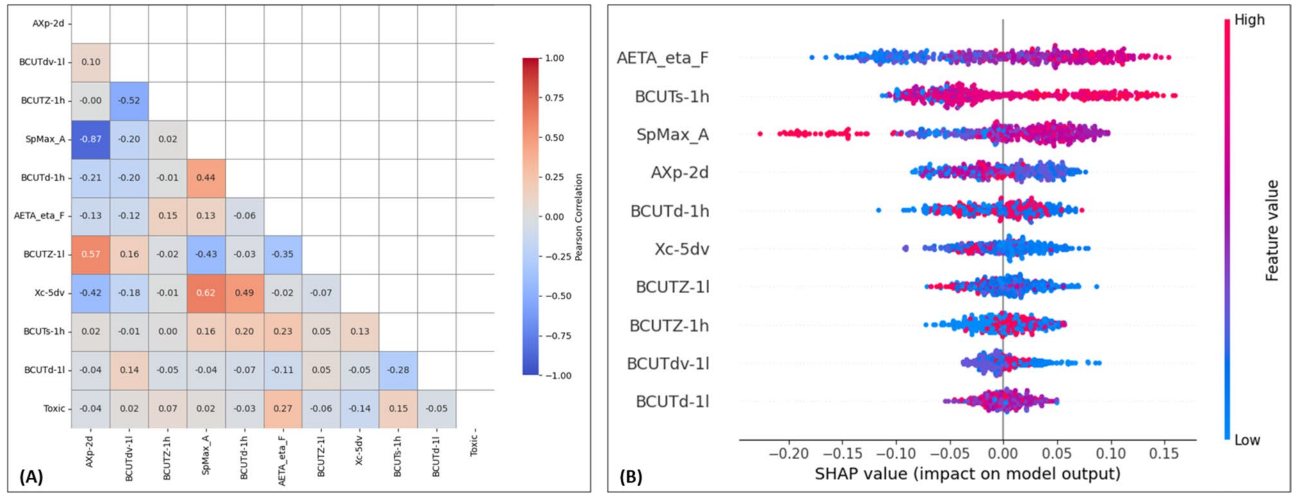

图3. (A) 所选描述符相关矩阵的热图。(B) 训练集每个描述符的SHAP值汇总图。

描述符	类型	定义
BCUTdv-1 L	BCUT 描述符	BCUT描述符来源于表示分子的邻接矩阵的特征值，并由特定属性加权（此处为范德华体积）。“1l”表示最低特征值。
BCUTZ-1 小时	BCUT 描述符	类似于 BCUTdv-1 L，但矩阵按原子极化率加权，“1 h”指的是最高特征值。
BCUTd-1 h	BCUT 描述符	BCUT 描述符，其中邻接矩阵根据偶极相关属性加权。“1 h”表示最高特征值。
BCUTZ-1 L	BCUT 描述符	一个由原子极化率加权的BCUT描述符，其中“1l”指最低特征值。
BCUTd-1 L	BCUT 描述符	一个由偶极相关性质加权的BCUT描述符，关注最低特征值。
BCUTs-1 小时	BCUT 描述符	一个由原子电负性加权的BCUT描述符，关注最高特征值。
AXp-2d	自相关描述符	这是一个二维自相关描述符，用于测量原子性质（如电负性、质量或 按分子中键距离加权的电荷。
Xc-5dv	自相关描述符	一种二维自相关描述符，考虑分子中价电子信息的分布。
SpMax_A	拓扑描述符	该描述符表示按原子性质加权的邻接矩阵的最大特征值。“¯A”后缀表示某种特定性质，例如电负性或极化率。
AETA_eta_F	与偶极相关的描述符	一种原子电负性加权的拓扑描述符。它特别与 F（氟）原子相关或 分子中的含氟特征。

表 2. 所选描述符的简要说明。

适用性领域 (AD) 分析

AD 对于评估[25,26]预测特定分子的结果不确定性非常重要。根据经济合作与发展组织（OECD）指南的原则 3[27],，在应用经过验证的模型来预测新数据点时，定义 AD 是必不可少的。只有当被分析的化合物落在 AD 范围内时，机器学习模型的可预测性才被认为是可靠的。在这里，杠杆法被用来定义 X-异常值（训练集）并识别位于 AD 之外的分子（在测试集的情况下）。在本研究中，描述符的数量为 10，训练样本数为 452，因此杠杆阈值为 0.066。结果表明，测试集化合物 200（杠杆值：0.229）、352（杠杆值：0.101）、171（杠杆值：0.101）、119（杠杆值：0.187）是异常值，因为这些化合物的杠杆值超过了阈值（0.066），如附录材料的图 S3B 所示。

通过 SHAP（Shapley 加法解释）图对描述符的解释

RFC模型识别了不同类型的描述符，例如BCUT描述符、自相关描述符、拓扑描述符、电负性和偶极相关描述符。BCUT描述符（例如，BCUTdv-1 L，BCUTZ-1 h）描述分子的大小、形状和电子性质，而自相关描述符（例如，AXp-2d，Xc-5dv）则提供了有关诸如电荷或电负性等性质在空间分布方面的见解（表2）。拓扑描述符（例如，SpMaxA）捕捉了连通性和结构特征。同时，电负性和偶极相关描述符（例如，AETAetaF）反映了分子的反应性和相互作用潜力，这对活性预测至关重要。

训练集描述符（AXp-2d、BCUTdv-1 L、BCUTZ-1 h、SpMax_A、BCUTd-1 h、AETA_eta_F、BCUTZ-1 L、Xc-5dv、BCUTs-1 h 和 BCUTd-1 L）的 SHAP（Shapley 加性解释；https://shap.readthedocs.io/en/latest/）总结图如图 3B 所示。在该图中，颜色从蓝色到红色的渐变

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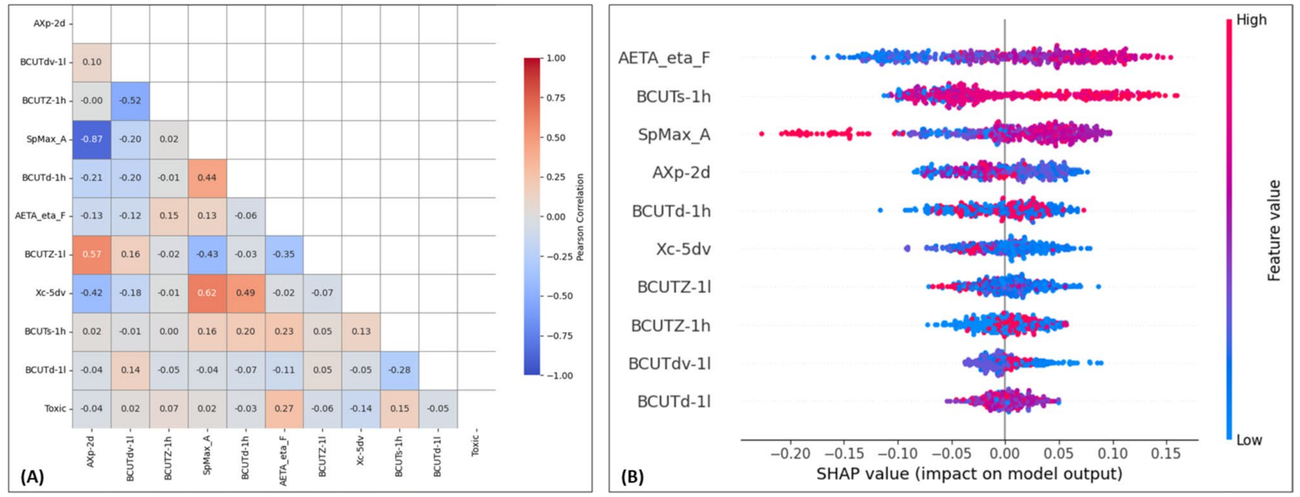

Fig. 3. (**A**) Heatmap of the correlation matrix of the selected descriptors. (**B**) Summary plot of SHAP values for each descriptor of the training set.

Descriptors	Types	Definition
BCUTdv-1 L	BCUT descriptors	A BCUT descriptor derived from eigenvalues of an adjacency matrix representing the molecule, weighted by a specific property (here, van der Waals volume). The "1l" indicates the lowest eigenvalue.
BCUTZ-1 h	BCUT descriptors	Similar to BCUTdv-1 L, but the matrix is weighted by atomic polarizability, and "1 h" refers to the highest eigenvalue.
BCUTd-1 h	BCUT descriptors	A BCUT descriptor where the adjacency matrix is weighted by a dipole-related property. "1 h" indicates the highest eigenvalue.
BCUTZ-1 L	BCUT descriptors	A BCUT descriptor weighted by atomic polarizability, where "1l" refers to the lowest eigenvalue.
BCUTd-1 L	BCUT descriptors	A BCUT descriptor weighted by dipole-related properties, focusing on the lowest eigenvalue.
BCUTs-1 h	BCUT descriptors	A BCUT descriptor weighted by atomic electronegativity, focusing on the highest eigenvalue.
AXp-2d	Autocorrelation descriptors	This is a 2D autocorrelation descriptor that measures the distribution of atomic properties (like electronegativity, mass, or charge) weighted by bond distance in a molecule.
Xc-5dv	Autocorrelation descriptors	A 2D autocorrelation descriptor that considers the distribution of valence electron information across a molecule.
SpMax_A	Topological descriptors	This descriptor represents the maximum eigenvalue of the adjacency matrix weighted by atomic properties. The "_A" suffix suggests a specific property, such as electronegativity or polarizability.
AETA_eta_F	Dipole-related descriptors	A measure of atomic electronegativity weighted topological descriptor. It specifically relates to the F (fluorine) atom or fluorine-related features in the molecule.

Table 2. A brief explanation of the selected descriptors.

Applicability domain (AD) analysis

The AD[25,26] is important for assessing the uncertainty in predicting a specific molecule. According to Principle 3 of the Organization for Economic Co-operation and Development (OECD) guidelines[27], it is essential to define the AD when applying validated models to predict new data points. The predictability of an ML model is considered reliable only if the compound being analyzed falls within the AD. Here, leverage approach is considered to define the X-outliers (training set) and identify the molecules that reside outside the AD (in the case of the test set). In this study, the number of descriptors is 10, the number of training samples is 452, hence the leverage threshold is found to be 0.066. The results suggest that the test set compounds 200 (Leverage: 0.229), 352 (Leverage: 0.101), 171 (Leverage: 0.101), 119 (Leverage: 0.187) are the outliers because the leverage values of these compounds are exceed the threshold value (0.066) as depicted in Figure S3B of the **Supplementary material**.

Interpretation of the descriptors through SHAP (SHapley additive exPlanations) plot

The RFC model identified different types of descriptors, such as BCUT descriptors, autocorrelation descriptors, topological descriptors, electronegativity and dipole-related descriptors. BCUT descriptors (e.g., *BCUTdv-1 L, BCUTZ-1 h*) describe molecular size, shape, and electronic properties, whereas autocorrelation descriptors (e.g., *AXp-2d, Xc-5dv*) offer insights into the spatial distribution of properties like charge or electronegativity[28] (Table 2). Topological descriptors (e.g., *SpMax_A*) capture connectivity and structural features. Meanwhile, electronegativity and dipole-related descriptors (e.g., *AETA_eta_F*) reflect molecular reactivity and interaction potential, crucial for activity predictions.

The SHAP (SHapley Additive exPlanations; https://shap.readthedocs.io/en/latest/) summary plot of the training set descriptors (*AXp-2d, BCUTdv-1 L, BCUTZ-1 h, SpMax_A, BCUTd-1 h, AETA_eta_F, BCUTZ-1 L, Xc-5dv, BCUTs-1 h*, and *BCUTd-1 L*) is presented in Fig. 3B. In this plot, the color gradient from blue to red

分别代表特征值从低到高。此颜色编码有助于解释每个描述符在RFC模型预测中的影响大小。例如，描述符AETA_eta_F显示正负SHAP值的广泛分布，表明其对模型结果的影响依赖于上下文，并受到与其他描述符相互作用的影响。一般来说，BCUTs-1 h的较高值（红色所示）与正的SHAP值相关，表明在分子某些区域中原子电负性增加倾向于推动预测向肾毒性方向发展（见图4）。相反，对于SpMax_A，较高的特征值（红色）通常与负的SHAP值相关，这表明具有整体较高拓扑极化性的分子，如克林霉素（化合物79），更可能被预测为“无毒”。此外，SHAP瀑布图（图4）提供了详细的

从图 4A 的瀑布图可以看出，AETA_eta_F 提供了最大的正向贡献，显著增加了所研究分子对“有毒”类别的预测。接下来的重要描述符 BCUTs-1 h 和 SpMax_A 也表现出良好的贡献，而 AXp-2d 和 BCUTZ-1 L 对兰索拉唑（化合物 366）的贡献较差。这种药物是一种质子泵抑制剂（PPI），用于治疗消化性溃疡、Zollinger-Ellison 综合征。它可能导致慢性肾毒性。同样地，从环丙沙星的瀑布图（图 4B）可以注意到，BCUTs-1 h、AETA_eta_F 和 SpMax_A 提供了最大的正向贡献，显著增加了对“有毒”类别的预测。它是一种常用的抗生素，已被发现与肾毒性相关。

另一方面，SpMax_A 提供了最大的负向贡献，显著增加了克林霉素（化合物 79，图 4C）和辛伐他汀（化合物 364，图 4D）向“非毒性”类别的预测。总体而言，这些观察结果表明，分子的特性如大小、形状、电子特性、芳香性以及诸如电荷或电负性等特征的空间分布在肾毒性中起着关键作用。这些特性可以在机制上与肾脏特异性毒性联系起来。例如，极性和偶极矩可能影响肾小管的积累和分泌，而芳香性可能促进生物活化生成反应性代谢物，从而在肾细胞中引起氧化应激或线粒体损伤。然而，肾毒性并不仅仅由这些分子属性决定；它还受包括剂量、暴露时间、个体患者特异性变量（例如遗传易感性、“p

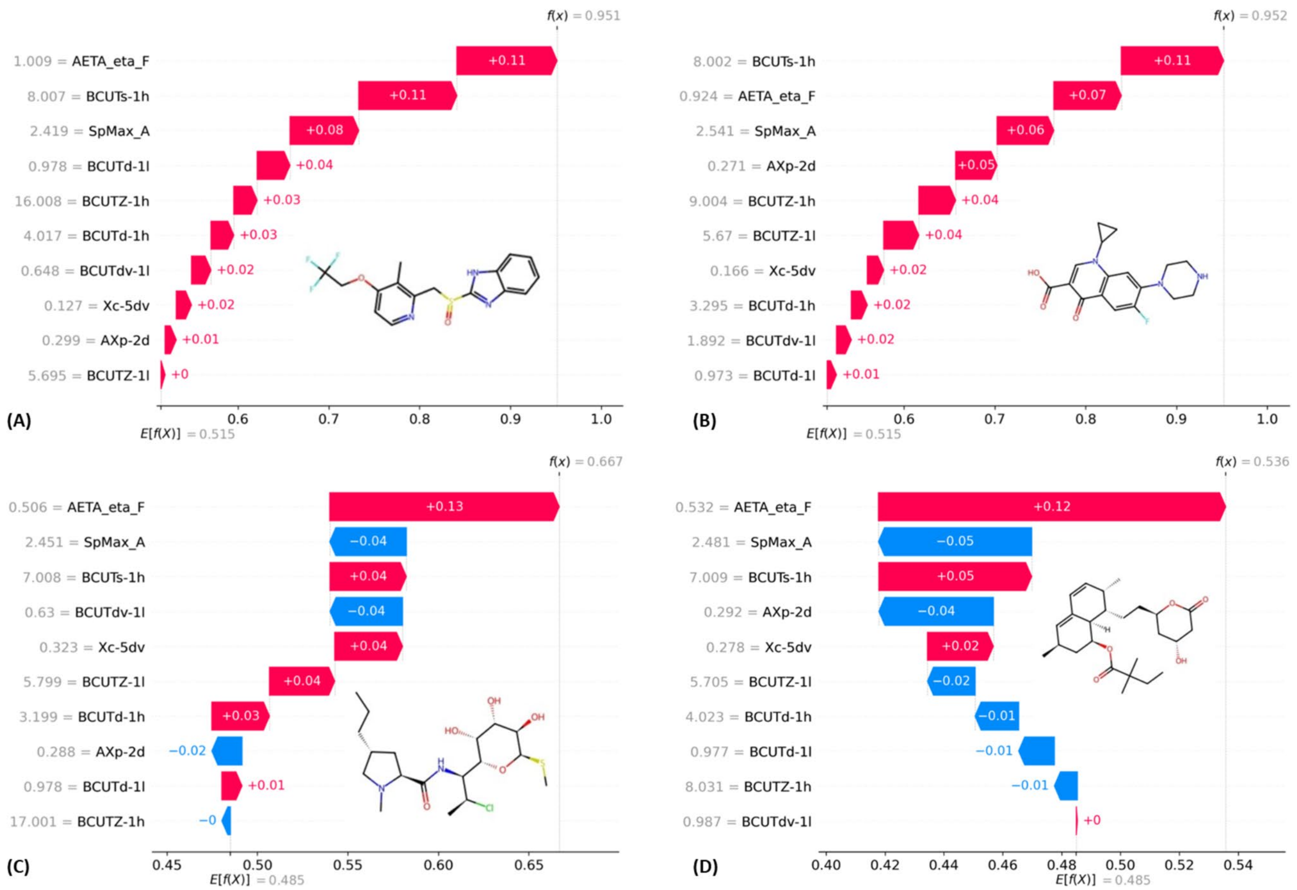

图4.（A）兰索拉唑（化合物366）、（B）环丙沙星（化合物400）、（C）克林霉素（化合物79）、（D）辛伐他汀（化合物364）的瀑布图。

represents low to high feature values, respectively. This color-coding helps to interpret the magnitude of each descriptor in the impact of the prediction of the RFC model. For example, the descriptor *AETA_eta_F* displays a broad distribution of both positive and negative SHAP values, indicating that its effect on the model outcome is context-dependent and influenced by interactions with other descriptors. In general, higher values of *BCUTs-1 h* (shown in red) are associated with positive SHAP values, indicating that increased atomic electronegativity in certain molecular regions tends to drive predictions toward kidney toxicity (see Fig. 4). In contrast, for *SpMax_A*, higher feature values (red) are often associated with negative SHAP values, suggesting that molecules with overall higher topological polarizability, such as Clindamycin (compound 79), are more likely to be predicted as 'Non-toxic'. Additionally, the SHAP waterfall plot (Fig. 4) provides a detailed breakdown of the contribution of individual features to the final prediction for a specific compound.

From the waterfall plot of Fig. 4A, it can be observed that the *AETA_eta_F* provides the largest positive contribution, significantly increasing the prediction toward the 'Toxic' class for the investigated molecules. The next important descriptors *BCUTs-1 h*, and *SpMax_A* exhibit good contributions, while the *AXp-2d*, *BCUTZ-1 L* having poor contributions for Lansoprazole (compound 366). This drug is a proton pump inhibitor (PPI) which is used in peptic ulcer disease, Zollinger-Ellison syndrome. It can lead to chronic kidney toxicity. Similarly, from the waterfall plot of Ciprofloxacin (Fig. 4B), it can be noticed that *BCUTs-1 h*, *AETA_eta_F*, and *SpMax_A* provide the largest positive contribution, significantly increasing the prediction toward the 'Toxic' class. It is a commonly used antibiotic that has been linked to nephrotoxicity.

On the other hand, *SpMax_A* provides the largest negative contribution, significantly increasing the prediction toward the 'Non-toxic' class for Clindamycin (compound 79, Fig. 4C), and Simvastatin (compound 364, Fig. 4D). Collectively, these observations indicate that molecular characteristics such as size, shape, electronic properties, aromaticity, and the spatial distribution of features like charge or electronegativity play a critical role in nephrotoxicity. These can be mechanistically linked to kidney-specific toxicity. For example, polarity and dipole moment may influence renal tubular accumulation and secretion, while aromaticity may promote bioactivation to reactive metabolites that induce oxidative stress or mitochondrial damage in kidney cells. However, kidney toxicity is not solely determined by these molecular attributes; it is also influenced by external factors including dosage, duration of exposure, individual patient-specific variables (e.g., genetic predisposition, pre-existing renal conditions), and potential drug-drug interactions.

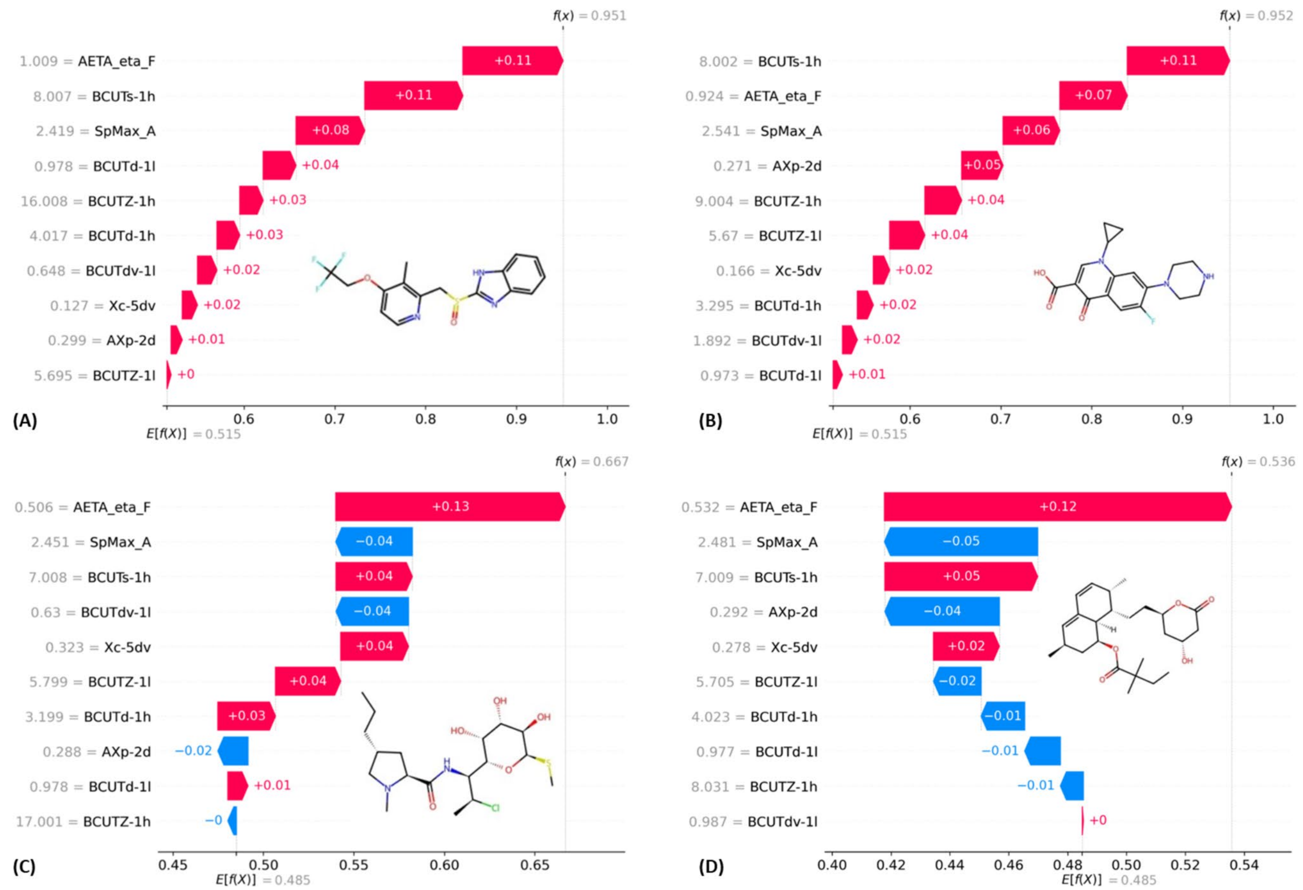

Fig. 4. Waterfall plots of the **(A)** Lansoprazole (compound 366), **(B)** Ciprofloxacin (compound 400), **(C)** Clindamycin (compound 79), **(D)** Simvastatin (compound 364).

qRASAR研究的描述符见解与可解释性

在本研究中，qRASAR 分类模型使用基于相似性和误差的指标在三种核函数下开发：欧几里得 (EUC)、高斯 (GK) 和拉普拉斯 (LK)[29]。每个模型结合了每个核下 10 个机器学习导出的描述符和 8 个 RASAR 描述符。基于重要性排序进行特征选择，以得出每个模型的简化版本，从而在不显著降低性能的情况下提高可解释性。特征重要性排序促使开发了简化模型，仅保留前三个描述符，通常为 MaxSim、MaxDist 和 Banerjee-Roy 系数。补充材料表 S3 中详细说明了 RASAR 描述符。我们已开发了 6 个基于分类的 qRASAR 模型，其中三个结合了机器学习导出的特征和基于误差的特征，而另外三个模型仅基于基于误差的特征开发。完整结果总结在表 S4 中。

考虑到特征数量和分类结果，EUC（已选）被选为六个开发模型中表现最好的模型。特征重要性分析在所有核中持续识别出三个关键描述符。EUC（已选）模型中的三个描述符是：

• MaxWtSim(EUC)：与类似物的最大加权相似性，结合了化学相似性和实验活性权重。• MaxNeg(EUC)：与无活性类似物的最大相似性，通过识别与已知无活性物质的结构接近性，作为防止假阳性的对策。• gm(EUC)（Banerjee–Roy 系数）：局部误差贡献的几何平均值，用于量化化学邻域中预测活性与观察活性之间的不一致性。

这些描述符不是随意的。它们是基于误差的表示，由机器学习选择的分子特征导出，这意味着最初由机器学习模型捕获的结构信息现在被编码为可解释的、化学意义明确的描述符[30]。这一转换使得模型在保持预测能力的同时提供透明性，这是监管应用和跨物种推理理由的基本要求。仅使用三个基于误差的描述符的模型能够优于更复杂模型的能力，说明了通过相似性加权和误差惩罚转换进行特征提炼的强大效果。这些描述符反映了基于类似体的预测的置信度和一致性，并且根植于已有的化学跨物种推理原则。qRASAR框架通过整合基于相似性的推理，相比标准QSAR提供了优势，这使得能够获得关于结构相关化合物为何共享特性的局部机理洞见

“KidneyTox_v1.0”：一个用于预测肾毒性的开放访问可解释人工智能平台

基于 Streamlit (https://streamlit.io) 的这个网页应用提供了一个多功能且稳健的框架，用于预测小分子可能的肾毒性风险（图 5）。基于严格验证的 AI/ML 模型，“KidneyTox_v1.0” (https://kidneytoxv1.streamlit.app/) 将查询化合物分为两类：‘有毒’或‘无毒’。该平台使用我们预训练的 ML 模型预测外部数据（查询分子）的结果，并通过 SHAP 分析提供预测的详细解释（图 6）。用户可以方便地输入 SMILES 字符串或在直观的 Python 网页应用界面中直接绘制查询分子的化学结构。用户还可以获得瀑布图，以便将其查询分子的预测肾毒性进行情境化。此外，该平台还提供了与查询分子最相似的数据集分子的可视化（基于 Tanimoto 相似性）。

在图6中，展示了通过“KidneyTox_v1.0”工具进行的示例预测。它提供了二维结构的可视化、预测结果（‘有毒’或‘无毒’）以及查询分子、与查询分子最相似的数据集分子的SHAP瀑布图。重要的是，这个用户友好的XAI平台有两个主要用途：

a. 促进可访问性和知识传播：通过将二维分子结构与查询化合物及已知肾毒性参考的 SHAP 瀑布图同时可视化，“KidneyTox_v1.0”平台能够提供关于特征对肾毒性预测贡献的可解释性洞察。同时，对于每个查询分子，该工具会根据与训练集的描述符计算其杠杆值。杠杆值低于定义阈值的分子被认为“在 AD 范围内”，表示预测可靠，而超过阈值的分子则为“在 AD 范围外”，提示需谨慎。此信息会与预测结果同时显示，“在 AD 范围内”以绿色显示，“在 AD 范围外”以红色显示。增强的可解释性有助于做出更明智的决策。

b. 为未来研究生成假设提供支持：该平台还通过提供来自描述符贡献的见解，影响新型“无毒”化学骨架的合理设计。这一能力是

Descriptor insights and interpretability of qRASAR study

In this study, qRASAR classification models were developed using descriptors derived from similarity and error-based metrics under three kernel functions: Euclidean (EUC), Gaussian (GK), and Laplacian (LK)[29]. Each model incorporated 10 ML-derived descriptors along with 8 RASAR descriptors per kernel. Feature selection was performed based on important rankings to derive simplified versions of each model, enhancing interpretability without significantly compromising performance. Feature importance rankings prompted the development of reduced models, preserving only the top three descriptors, typically MaxSim, MaxDist, and the Banerjee-Roy coefficient. Table S3 of the Supplementary material highlighted the detailed description of RASAR descriptors. We have developed 6 classification-based qRASAR models where three are combining ML-derived features and error-based features, while 3 models are developed based on only error-based features. Complete results are summarized in Table S4 of the Supplementary material. Among all selected (reduced) models, the EUC (Selected) model outperformed its GK and LK counterparts, achieving a test accuracy of 0.7568, precision of 0.7400, recall of 0.7255, and an F1-score of 0.7327. This indicates a strong balance between sensitivity and specificity and establishes EUC (Selected) as the best-performing model, despite using only three error-based descriptors. When descriptors were reduced to the top three features in the selected models, there was a slight drop in performance across all kernels. For instance, the LK-selected model exhibited an F1-score of 0.6535 compared to 0.7475 in the full model. This drop highlights the contribution of additional descriptors to model robustness, even though the selected models maintained reasonable accuracy.

Considering the number of features and classification result, EUC (Selected) has been selected as the best model among all six developed models. Feature importance analysis consistently identified three key descriptors across all kernels. The three descriptors in the EUC (Selected) model were:

- *MaxWtSim(EUC)*: The maximum weighted similarity to analogs, incorporating both chemical similarity and the weight of experimental activity.
- *MaxNeg(EUC)*: The maximum similarity to inactive analogs, acting as a countermeasure to false positives by identifying structural proximity to known inactives.
- *gm(EUC) (Banerjee–Roy coefficient)*: A geometric mean of local error contributions, quantifying inconsistency between predicted and observed activities in the chemical neighborhood.

These descriptors are not arbitrary. They are error-based representations derived from ML-selected molecular features, meaning the structural information originally captured by the ML model is now encoded into interpretable, chemically meaningful descriptors[30]. This transformation enables the model to retain predictive strength while offering transparency, an essential requirement for regulatory applications and read-across justifications. The ability of a model with just three error-based descriptors to outperform more complex models illustrates the power of feature distillation through similarity-weighted and error-penalized transformations. These descriptors reflect both the confidence and consistency of predictions based on analogs and are rooted in established chemical read-across principles. The qRASAR framework provides an advantage over standard QSAR by integrating similarity-based reasoning, which allows local mechanistic insights into why structurally related compounds share nephrotoxic outcomes. In contrast to black-box deep learning, which may offer high accuracy but limited transparency, qRASAR predictions are chemically interpretable and directly actionable, linking descriptors and scaffolds to toxicity mechanisms.

“KidneyTox_v1.0”: an open-access XAI platform for predicting nephrotoxicity

This Streamlit (https://streamlit.io)-based web application provides a versatile and robust framework for predicting the potential risk of nephrotoxicity associated with small molecules (Fig. 5). Based on a rigorously validated AI/ML model, the “KidneyTox_v1.0” (https://kidneytoxv1.streamlit.app/) classifies query compounds into two clear categories: ‘Toxic’ or ‘Non-toxic’. This platform predicts the outcomes for external data (query molecules) using our pre-trained ML model and provides a detailed explanation of the predictions using SHAP analysis (Fig. 6). Users can conveniently input the SMILES string or directly draw the chemical structure of the query molecule within an intuitive, Python-based web application interface. Users also receive waterfall plots to contextualize the predicted nephrotoxicity of their query molecules. In addition, this platform offers the visualization of the most similar molecule from the dataset concerning the query molecule (based on the Tanimoto-similarity). The employed dataset and codes are available on GitHub (https://github.com/Aminchem inform/KidneyTox_v1.0).

In Fig. 6, an example prediction *via* “KidneyTox_v1.0” tool is depicted. It offers the visualization of 2D structures, predicted results (either ‘Toxic’ or ‘Non-toxic’), and the SHAP waterfall plot of the query molecule, the most similar molecule from the dataset to the query molecule. Importantly, this user-friendly XAI platform serves two primary purposes:

a. *Facilitating accessibility and knowledge dissemination*: By visualizing 2D molecular structures alongside SHAP waterfall plots for both query compounds and known kidney toxic references, the “KidneyTox_v1.0” platform provides interpretable insights into feature contributions toward nephrotoxicity predictions. Meanwhile, for each query molecule, this tool calculates the leverage value based on the descriptors relative to the training set. Molecules with leverage below the defined threshold are considered “Within AD”, indicating reliable predictions, while molecules above the threshold are “Outside AD”, signalling caution. This information is displayed alongside the prediction, with “Within AD” shown in green and “Outside AD” in red. This enhanced interpretability supports more informed decision-making.
b. *Enabling hypothesis generation for future research*: This platform also influences the rational design of novel ‘Non-toxic’ chemical scaffolds by providing insights from descriptor contributions. This capability is par-

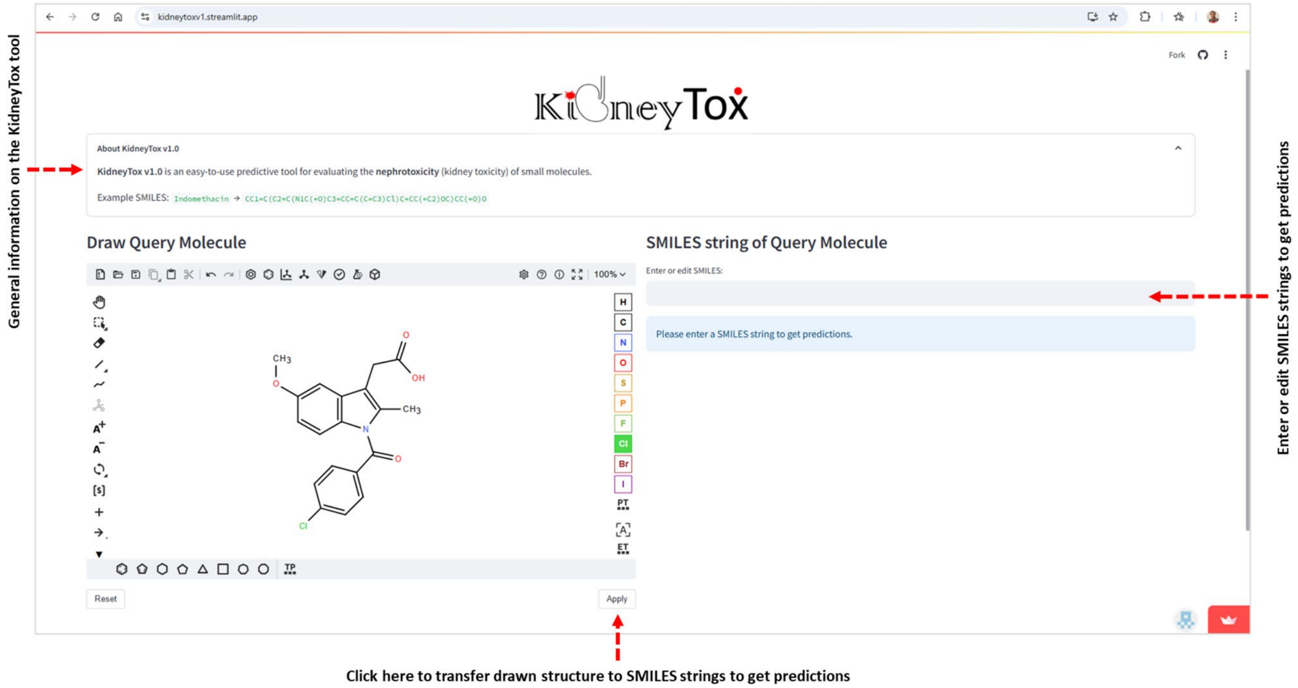

图5.“KidneyTox_v1.0”工具的界面。实际输入是SMILES字符串。查询分子可以直接以SMILES格式粘贴或输入，也可以通过分子绘图器插入。如果用户直接粘贴、输入或编辑了SMILES字符串，则可以通过按“Enter”键开始计算/预测。如果用户使用分子绘图器，则可以通过点击“应用”按钮开始预测。

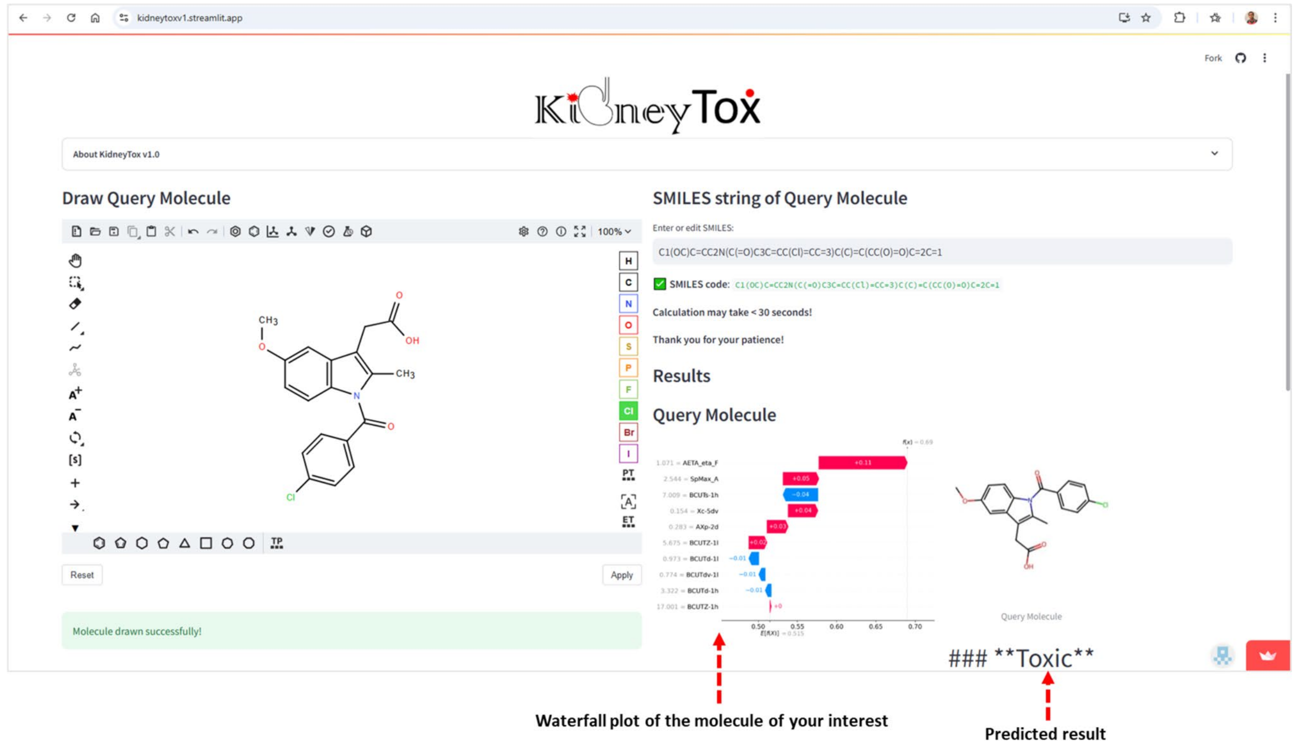

图6. 查询分子的二维结构可视化、预测结果（“有毒”或“无毒”）、适用性域分析结果以及SHAP瀑布图，以及相对于查询分子在数据集中最相似的分子。面板顶部为查询分子及向上箭头按钮，用于滚动到页面顶部。用户可以下载图表和二维结构。

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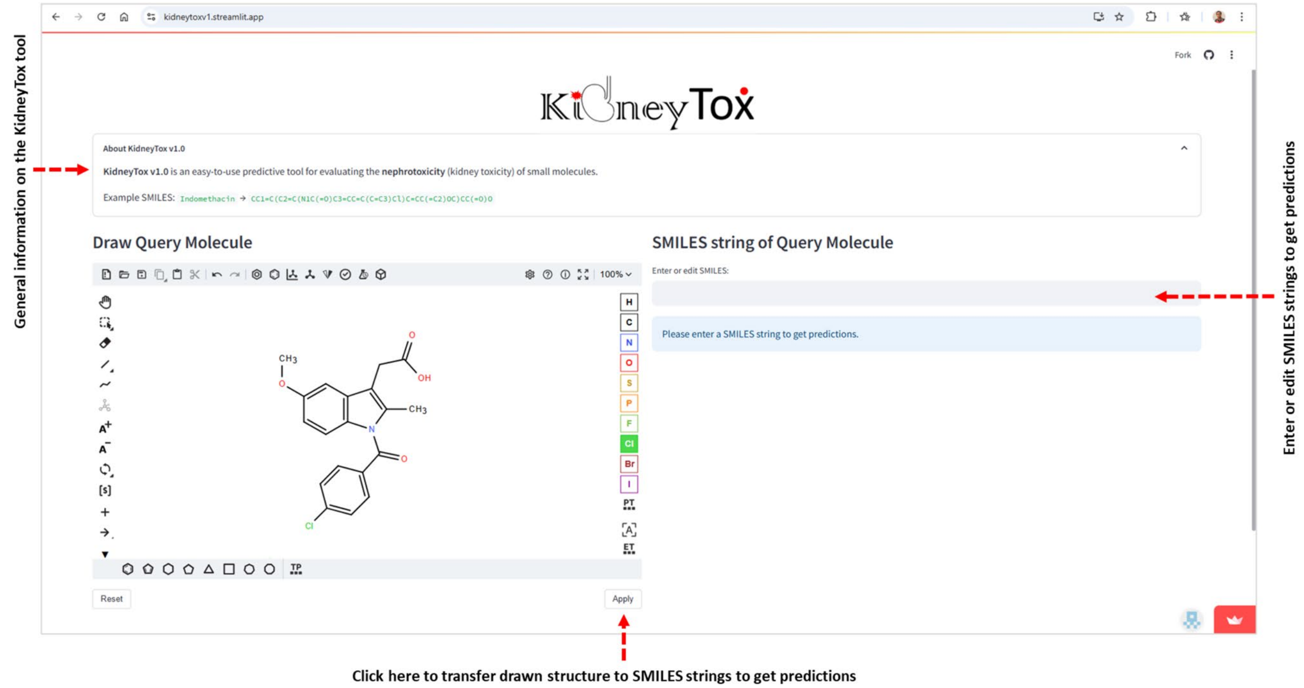

Fig. 5. The interface of "KidneyTox_v1.0" tool. The actual input is a SMILES string. Query molecules can be directly pasted or typed in SMILES format or inserted through the molecular sketcher. If user directly pasted or typed or Edited in SMILES strings, then the calculations/predictions can be started by pressing "Enter". If the user uses the molecular sketcher, then the predictions can be started by clicking on the "Apply" button.

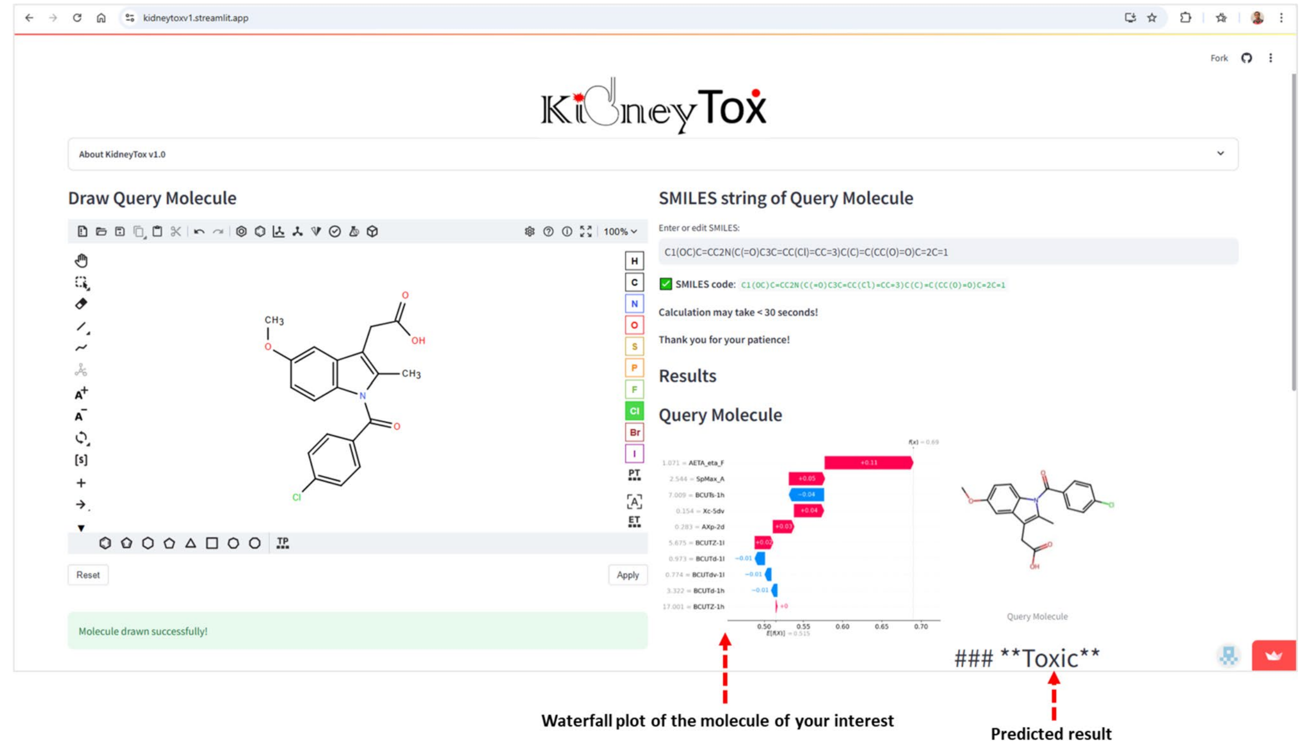

Fig. 6. The visualization of 2D structures, predicted results (either 'Toxic' or 'Non-toxic'), applicability domain analysis results, and the SHAP waterfall plots of the query molecule, and the most similar molecule from the dataset with respect to the query molecule. The panel is headed by the query molecule and an up-arrow button to scroll to the top of the page. Users can download the plot(s) and the 2D structure(s).

在推进肾毒性预测的药物发现工作中尤其有价值，特别是在缺乏大规模筛选数据的情况下。

讨论与结论

本研究提出了一个整合的化学信息学和人工智能/机器学习框架，用于利用包含565种化合物的化学多样性数据集预测药物引起的肾毒性。该数据集表现出多样的化学空间，其中大多数分子为中等亲脂性（平均LogP为=1.81），包含一到两个芳香环，~7个可旋转键，~6个氢键受体，以及~3个氢键供体。平均拓扑极性表面积（TPSA）和分子量（MW）分别为113.5 Å² 和416.9，极端值从高度亲水到高度亲脂化合物不等。因此，全面的化学空间分析显示了分子性质的广泛分布，突出了肾毒性和非毒性药物的结构和理化多样性。此外，使用“Fasda_v1.0”的骨架多样性分析证实了五个簇中高度的唯一性。簇0显示出最高的多样性，包含149种化合物、126个Bemis-Murcko骨架，以及0.857的单例比率。簇1-4同样显示出高度多样性。

我们使用所选的Mordred描述符开发了一个经过Optuna优化的RFC模型（最佳参数：n_estimators = 60，max_depth = 24，min_samples_split = 13，min_samples_leaf = 2），并取得了出色的性能（准确率 = 0.841，精确率 = 0 .830，召回率 = 0.830，F1 = 0.830）。基于SHAP的可解释人工智能（XAI）分析确定了关键描述符（如AETA_eta_F, BCUTs-1 h, SpMax_A）对毒性预测有显著贡献，其可解释性趋势将电负性、柔性和极化性与肾毒性结果联系起来。值得注意的是，瀑布图展示了描述符对已知肾毒性药物（如兰索拉唑、环丙沙星）和非毒性药物（如辛伐他汀）的特定影响，确认了模型的可解释性。同时，已识别的异常值如化合物119（一个非常小且高极性的羧酸盐）、171（一个紧凑的N–C–C–S片段，具有非典型电负性模式）、200（一个富含氮的多环芳香系统，具有高

最终，这些努力 culminated 于“KidneyTox_v1.0”的部署，这是一个开放访问的、集成了 SHAP 的肾毒性预测网络平台。该工具通过为任何查询分子提供可视化、可解释的预测，支持假设生成、化合物优先级排序（例如，减少过于电子丰富的芳香系统、调节极性、调整柔性）以及早期药物发现中的骨架优化。重要的是，“KidneyTox_v1.0”在设计上注重可用性和可持续性。随着新的肾毒性数据的出现，该工具将定期更新，并且用户可以以标准 SMILES 格式上传专有化合物。上传的数据在内存中处理而不被存储，以确保用户隐私和数据匿名性。

方法

数据集

药物诱导的肾毒性（N_{Total} = 565）数据集来源于文献[17,18]。表S1提供了所选肾毒性药物及其相关毒性特征的概览。该数据集包含565种化学结构多样的小分子，其中包括287种报道在人类中可引起肾毒性的药物（‘有毒’）和278种已知无肾毒性作用的药物（‘无毒’）。

化学空间分析

计算了六个常见的物化描述符，即辛醇-水分配系数（LogP）、分子量（MW）、氢键受体（HBA）、氢键供体（HBD）、分子中环系统的总数（nRings）以及可旋转键的数量（nRB），使用“RDKit”（https://www.rdkit.org/）化学信息学工具包（版本 rdkit-pypi-2022.9.5）[31]。然后绘制了箱形图以理解所研究分子的化学空间。

之后，使用“Fasda_v1.0”工具（https://github.com/Amincheminfom/Fasda_v1）对这些与药物性肾损伤相关的研究化合物进行了指纹辅助支架多样性分析。这种指纹辅助支架多样性分析方法结合了分子指纹、相似性分析、降维和聚类，以理解分子数据集中的结构多样性和关系。此外，将分子划分为不同的簇有助于对结构相似的化合物进行分类，这可能对支架分析等任务有用[19]。

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ticularly valuable for advancing drug discovery efforts in nephrotoxicity prediction, especially in contexts lacking large-scale screening data.

Discussion & conclusion

This study presents an integrative cheminformatics and AI/ML framework for predicting drug-induced nephrotoxicity using a chemically diverse dataset of 565 compounds. The dataset exhibits diverse chemical space, with most molecules being moderately lipophilic (average *LogP* = 1.81), containing one or two aromatic rings, ~7 rotatable bonds, ~6 hydrogen bond acceptors, and ~3 hydrogen bond donors. Average *TPSA* and *MW* are 113.5 Å² and 416.9, respectively, with extremes ranging from highly hydrophilic to highly lipophilic compounds. Hence, comprehensive chemical space analysis revealed a broad distribution of molecular properties, underscoring the structural and physicochemical diversity of nephrotoxic and non-toxic drugs. Moreover, scaffold diversity analysis using “Fasda_v1.0” confirmed high uniqueness across five clusters. Cluster 0 shows the highest diversity with 149 compounds, 126 Bemis-Murcko scaffolds, and a singleton ratio of 0.857. Clusters 1–4 also display high diversity, with singleton ratios ranging from 71.9% to 89.9%. Overall, all clusters carry more than 70% singleton scaffolds, underscoring the high structural diversity across the dataset.

We developed an Optuna-optimized RFC model (best parameters: *n_estimators* = 60, *max_depth* = 24, *min_samples_split* = 13, *min_samples_leaf* = 2) using selected Mordred descriptors, achieving excellent performance (Accuracy = 0.841, Precision = 0.830, Recall = 0.830, F1 = 0.830). SHAP-based XAI analysis identified key descriptors (e.g., *AETA_eta_F*, *BCUTs-1 h*, *SpMax_A*) as significant contributors to toxicity prediction, with interpretable trends linking electronegativity, flexibility, and polarizability to nephrotoxicity outcomes. Notably, waterfall plots illustrated descriptor-specific impacts on known nephrotoxic (e.g., Lansoprazole, Ciprofloxacin) and non-toxic (e.g., Simvastatin) drugs, confirming model interpretability. Meanwhile, the identified outliers such as compounds 119 (a very small and highly polar carboxylate), 171 (a compact N–C–C–S fragment with atypical electronegativity patterns), 200 (an N-rich polycyclic aromatic system with high conjugation and polarizability), and 352 (a large and structurally complex peptidic boronic-acid scaffold) deviate markedly from the physicochemical space of the dataset. Consistent with this, these outliers (compounds 119, 171, 200, and 352) exhibit extreme values in key descriptors (Table S5 of the Supplementary material). For example, unusually low or high *BCUT* indices, atypical atomic charge distributions (*AXp-2d*, *AETA_eta_F*), and divergent topological indices (*SpMax_A*, *Xc-5dv*) reflecting structural and electronic profiles that differ markedly from the main dataset. Overall, these findings indicate that molecular properties such as size, shape, electronic distribution, aromaticity, and the spatial arrangement of electronegative or polar groups play critical roles in nephrotoxicity. These properties can be mechanistically linked to kidney-specific toxic outcomes. For instance, polarity and dipole moment can influence renal tubular uptake and accumulation, while aromaticity may facilitate bioactivation to reactive metabolites associated with mitochondrial dysfunction or oxidative stress in kidney cells. These insights provide actionable guidance for optimizing molecular structures to reduce predicted nephrotoxic risk. In parallel, qRASAR models were developed using similarity and error-based descriptors. Among all models, the reduced EUC (Selected) model, using only three interpretable, error-based descriptors, offered a strong balance of accuracy (F1-score: 0.7327) and interpretability, emphasizing the utility of transparent models in regulatory contexts. The dataset size may represent a limitation of the present study. Future investigations could strengthen external validation by incorporating larger datasets (e.g., Tox21, PubChem bioassays data).

Finally, these efforts culminated in the deployment of “KidneyTox_v1.0”, an open-access, SHAP-integrated web platform for nephrotoxicity prediction. This tool supports hypothesis generation, compound prioritization (e.g., reducing overly electron-rich aromatic systems, modulating polarity, adjusting flexibility), and scaffold optimization in early drug discovery by offering visual, interpretable predictions for any query molecule. Importantly, “KidneyTox_v1.0” is designed for usability and sustainability. This tool will be updated periodically as new nephrotoxicity data become available, and users can upload proprietary compounds in standard SMILES formats. Uploaded data are processed in memory without being stored, ensuring user privacy and data anonymization.

Methods

Dataset

Drug-induced nephrotoxicity (N_{Total} = 565) dataset was sourced from literature[17,18]. Table S1 provides an overview of the selected nephrotoxic drugs and their associated toxicity profiles. The dataset comprises 565 chemically diverse small molecules, including 287 drugs reported to induce nephrotoxicity in humans (‘Toxic’) and 278 drugs with no known nephrotoxic effects (‘Non-toxic’).

Chemical space analysis

Six common physicochemical descriptors namely octanol-water partition coefficient (*LogP*), molecular weight (*MW*), hydrogen bond acceptors (*HBA*), hydrogen bond donors (*HBD*), total number of ring systems in the molecule (*nRings*), number of rotatable bonds (*nRB*) were calculated by using the “RDKit” (https://www.rdkit.org/) cheminformatics toolkit (version rdkit-pypi-2022.9.5)[31]. Then bins plots were drawn to understand the chemical space of the investigated molecules.

After that “Fasda_v1.0” tool (https://github.com/Amincheminfom/Fasda_v1) was used to perform fingerprint-assisted scaffold diversity analysis of these investigated compounds associated with drug-induced nephrotoxicity. This fingerprint-assisted scaffold diversity analysis approach combines molecular fingerprinting, similarity analysis, dimensionality reduction, and clustering to understand the structural diversity and relationships within a dataset of molecules. Moreover, dividing the molecules into clusters helps in categorizing structurally similar compounds, which may be useful for tasks like scaffold analysis[19].

机器学习模型开发

Mordred 描述符是使用内部 “Fiore_v1.0” 平台（https://github.com/Amincheminfom/Fiore_v1.0）计算的。包含缺失值、非数字条目或准恒定行为（定义为单个值在超过 98% 的样本中出现）的特征被排除在进一步分析之外。其目标是清理数据集并通过仅保留最具信息量的描述符来减少特征空间，从而提高模型性能、可解释性和计算效率。

RFC 被使用，并使用 Optuna (https://optuna.org/) 进行了超参数优化。以下 RFC 超参数在指定范围内进行了调优：(i) 估算器数量 (n_estimators)：10–100；(ii) 最大树深度 (max_depth)：2–32；(iii) 拆分内部节点所需的最小样本数 (min_samples_split)：2–16；以及 (iv) 叶节点所需的最小样本数 (min_samples_leaf)：1–16。所得 RFC 模型使用前述的统计性能指标进行了评估[21]。

跨读特征计算

一旦使用分子描述符和指纹开发了AI-ML QSAR模型，就使用相同的特征空间生成基于相似性和误差的RASAR描述符。首先通过将训练集拆分为子训练集和验证集，并使用Read-Across-v4.2.2工具[32]生成验证预测，来确定最优的Read-Across超参数。紧密同源体的数量（2–10）被系统地变化，并根据验证性能选择最优设置。然后使用这些优化参数，通过RASAR-Desc-Calc-v3.0.3工具[32]计算训练集和测试集的RASAR描述符。计算的RASAR描述符完整列表见补充材料的表S2。

像QSAR分析一样，对RASAR描述符矩阵也进行了特征选择，以识别最具区分力的特征。然而，在此之前，我们特意移除了RASAR描述符SD_Activity、SE和CVact，它们分别代表近似同系物活性值的加权标准差、相应的标准误差以及变异系数[29]。我们采用了特征选择算法，即识别最具区分力的特征，这一算法由于其与建模算法无关的特性，旨在实现公正的特征选择。

qRASAR 模型开发与验证

qRASAR 分类模型是使用在三种核函数下计算的基于相似性和误差的跨读描述符构建的：EUC、GK 和 LK。对于每种核函数，开发了两类模型：(i) 混合模型，将机器学习选择的分子描述符与核特定的 RASAR 描述符结合；(ii) 仅 RASAR 模型，仅使用误差/相似性描述符。这样总共生成了六个分类模型（EUC-Hybrid、EUC-RASAR、GK-Hybrid、GK-RASAR、LK-Hybrid、LK-RASAR）。外部性能通过保留的测试集评估，并使用准确率（Accuracy）、精确率（Precision）、召回率（Recall）和 F1 值。为了增强可解释性，应用了特征重要性排序，并通过保留排名前三的基于误差的描述符获得了精简（“Selected”）模型。在所有核函数中，有三个描述符被一致优先考虑：MaxWtSim（与类似物的最大加权相似性）、MaxNeg（与已知非活性物质的最大相似性）以及 gm（Banerjee-Roy 系数），它捕捉了邻域不一致性

数据可用性

数据 a 在手稿、补充材料以及 GitHub 页面 (h t t p s : / / g i t h u b . c o m / A m) 上可用
英寸 m i n f o r m / 肾脏毒性_ v 1.0) 。

接收日期：2025年7月14日；接受日期：2026年1月6日
Published online: 13 January 2026

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Machine learning model development

Mordred descriptors[20] were calculated by using *in house* “Fiore_v1.0” platform (https://github.com/Amincheminfom/Fiore_v1.0). Features containing missing values, non-numeric entries, or quasi-constant behavior (defined as a single value appearing in more than 98% of samples) were excluded from further analysis. The objective was to clean the dataset and reduce the feature space by retaining only the most informative descriptors, thereby enhancing model performance, interpretability, and computational efficiency.

RFC were employed, with hyperparameter optimization conducted using Optuna (https://optuna.org/). The following RFC hyperparameters were tuned within the specified ranges: (i) number of estimators (*n_estimators*): 10–100; (ii) maximum tree depth (*max_depth*): 2–32; (iii) minimum number of samples required to split an internal node (*min_samples_split*): 2–16; and (iv) minimum number of samples required at a leaf node (*min_samples_leaf*): 1–16. The resulting RFC model were evaluated using statistical performance metrics as described earlier[21].

Read-across feature calculation

Once the AI-ML QSAR models were developed using molecular descriptors and fingerprints, the same feature spaces were used to generate similarity- and error-based RASAR descriptors. Optimal Read-Across hyperparameters were first identified by splitting the training set into sub-training and validation sets and generating validation predictions using Read-Across-v4.2.2 tool[32]. The number of close congeners (2–10) was systematically varied, and the optimal setting was selected based on validation performance. These optimized parameters were then used to compute RASAR descriptors for both training and test sets using RASAR-Desc-Calc-v3.0.3 tool[32]. The complete list of computed RASAR descriptors is provided in Table S2 of the Supplementary Material.

Like the QSAR analysis, feature selection was performed on the RASAR descriptor matrix to identify the most discriminating features. However, before this, we have deliberately removed the RASAR descriptors SD_Activity, SE, and CVact, which stands for the weighted standard deviation of the activity values of the close congeners, the corresponding standard error, and the coefficient of variation[29]. We employed the feature selection algorithm, i.e., identifying the most discriminating features, which aimed towards an unbiased feature selection due to its modeling algorithm-independent nature.

qRASAR model development and validation

qRASAR classification models were constructed using similarity- and error-based read-across descriptors computed under three kernels: EUC, GK, and LK. For each kernel, two model families were developed: (i) Hybrid models combining the ML-selected molecular descriptors with kernel-specific RASAR descriptors, and (ii) RASAR-only models using only the error/similarity descriptors. This yielded six classification models in total (EUC-Hybrid, EUC-RASAR, GK-Hybrid, GK-RASAR, LK-Hybrid, LK-RASAR). External performance was assessed on the held-out test set using Accuracy, Precision, Recall, and F1. To enhance interpretability, feature importance ranking was applied and reduced (“Selected”) models were derived by retaining the top three error-based descriptors. Across kernels, three descriptors were consistently prioritized: MaxWtSim (maximum weighted similarity to analogs), MaxNeg (maximum similarity to known inactives), and gm (Banerjee-Roy coefficient), which captures neighborhood inconsistency via an error-weighted geometric mean. These error-aware, similarity-weighted descriptors distill the structural signal captured by the ML stage into chemically interpretable quantities. Among all six models, the EUC-Selected model showed the best overall performance, striking a strong balance between sensitivity and specificity while maintaining interpretability.

Data availability

Data are available in the manuscript, Supplementary material, and on the GitHub page (https://github.com/Amincheminform/KidneyTox_v1.0).

Received: 14 July 2025; Accepted: 6 January 2026
Published online: 13 January 2026

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致谢

SK 感谢 Kean 大学 Dorothy 和 George Hennings 科学、数学与技术学院（HCSMT）的管理层，通过释放时间和提供资源为研究提供机会。SAA 和 SP 真诚感谢意大利萨莱诺大学提供研究设施。

作者贡献

Sk Abdul Amin：概念化、数据整理、正式分析、方法学、撰写——原始草稿、审阅与编辑。Supratik Kar：概念化、正式分析、资源、监督、撰写——原始草稿、审阅与编辑。Stefano Piotto：资源、监督、撰写——审阅与编辑。

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Acknowledgements

SK would like to thank the administration of the Dorothy and George Hennings College of Science, Mathematics, and Technology (HCSMT) at Kean University for providing research opportunities through release time and resources. SAA and SP sincerely acknowledge the University of Salerno, Italy for providing the research facilities.

Author contributions

Sk Abdul Amin: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, review & editing. Supratik Kar: Conceptualization, Formal analysis, Resources, Supervision, Writing – original draft, review & editing. Stefano Piotto: Resources, Supervision, Writing – review & editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-35496-4.

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